Online Informative Path Planning for Dynamic Field Mapping with a Learned Prior and GP Residual Uncertainty

Course: Mobile Robotics

Shayesteh Hafezi

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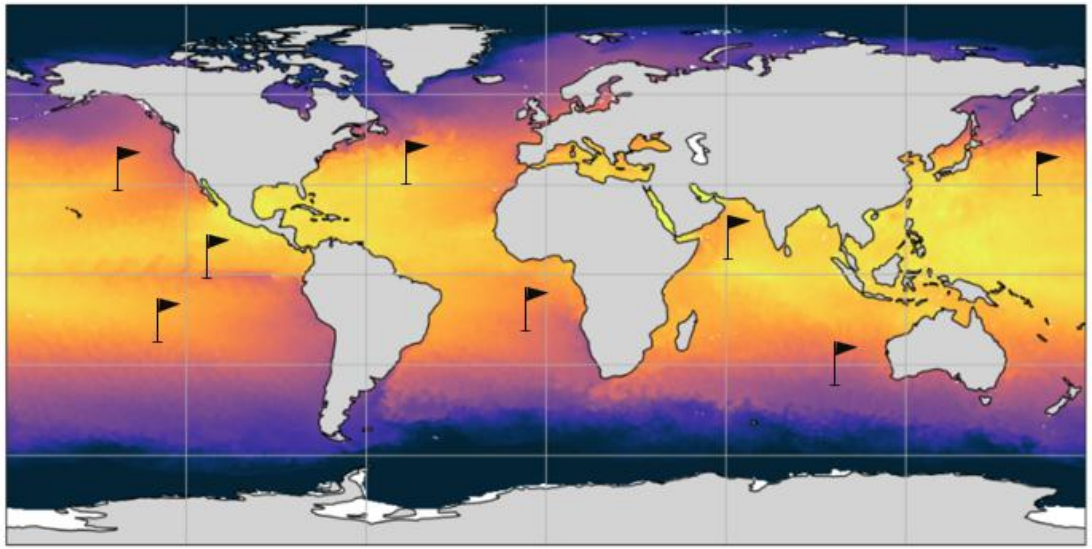
Online IPP for Dynamic Field Mapping

Problem

Given a time-varying scalar field $f(x, t)$ and noisy samples along robot trajectories,
maintain a belief (mean + uncertainty) and plan motions that maximize information gained per travel cost in real time.

Key idea / contribution

Use a fast dynamic prior to predict the field, then learn a probabilistic GP residual to capture where the prior is wrong.
Drive IPP using residual mean $|\mu_e|$ and σ_e (uncertainty) with travel/feasibility cost.



Method: Predict → Update → Plan → Move

A loop for dynamic mapping

Predict
Dynamics prior

→

Measure
Noisy samples
$y = f + \varepsilon$

→

Update
GP on residual
$e = y - f_{prior}$

→

Plan
Argmax
$(\alpha|\mu_e| + \beta\sigma_e) - \lambda\, cost$

→

Move
Execute

Belief representation

Field estimate:

$$\hat{f}(x, t_k) = \hat{f}_{prior}(x, t_k) + \mu_e(x)$$

Uncertainty proxy:

$$Var[f(x, t_k)] \approx \sigma_e^2(x)$$

Residual GP is updated online each assimilation step, enabling uncertainty-driven planning.

Planning objective (IPP)

Choose targets that balance:

- Exploration: high σ_e (unknown)
- Exploitation: high $|\mu_e|$ (biased prior)
- Motion cost: distance / feasibility

Stretch: multi-robot via Voronoi cells + same acquisition per cell.

Evaluation plan + timeline

What I will build, compare, and deliver

Experimental setup

- 2D dynamic field with ground-truth simulator
- Noisy point measurements along trajectories
- Primary: single robot; Stretch: 2–4 robots

Baselines + metrics

Baselines:
- Random / lawnmower
- GP max-variance (no dynamic prior)
- Prior-only (no GP residual)

Metrics:
- Grid RMSE vs time
- $\int \sigma^2(x)dx$ (uncertainty)
- RMSE reduction per distance

Timeline (Feb 23 → Apr 21) + deliverables

Early Mar: simulator + sensing + baselines. Mid Mar: integrate predict–update residual-GP mapping.
Late Mar: implement IPP controller + main experiments/ablations. Early Apr → deadline: learned prior (stretch: FNO), finalize results, public Git repo + 7-min video + 6-page IEEE-style report + poster.